

gRPC MACH Architecture

TCO Costs Efficiency Calculator

Point-to-Point Integration Pattern · Protocol Buffer IDL · AWS Egress Pricing

✔ File 'unaryRPCPattern.proto' uploaded successfully!

A7 **Activity 7: Calculate the TCO Cost Considering Optimisation Strategies**

The Phase 4 report reflects the full optimised TCO: networking egress + cloud infrastructure – FinOps RI/SP savings. Unit economics, ARPU, ROI, break-even users, and the stakeholder PDF report are all generated here.



Phase 4 — TCO Networking Costs Report

Full cost breakdown including architecture tactics, unit economics, and stakeholder summary.

SERVICE UNDER ANALYSIS

order-retriever-system

pnomom

BUC1

1000 base RPS

350 End Users

–

APPLIED ARCHITECTURE TACTICS

Server-side LB Timeout Retry Circuit Breaker mTLS OAuth 2.0 + JWT

Applied Architecture Tactics & Patterns

INFORMATIONAL — NO RPS IMPACT

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Server-side Load Balancing	–	Balances the reception of gRPC packets across backend replicas.	None

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Timeout	0 ms	Discards lost packets and frees the client thread after the configured wait.	None
Circuit Breaker	Min calls: 0 Half-open calls: 0 Wait: 0 ms Failure threshold: 0%	Opens the circuit when failure thresholds are exceeded, protecting downstream services.	None
mTLS (mutual TLS)	0 reconnects/hr	Both client and server present X.509 certificates. CA validates both. 30: Max Payload overhead: bytes per message — RFC 8446 Because of RFC 8446 Section-5.2 * TLSCiphertext outer header: 5 bytes: * + 1-byte opaque_type=23 (uint8 ContentType) * + 2-byte legacy_record_version (uint16 ProtocolVersion) * + 2-byte length (uint16 length). * TLSInnerPlaintext prefix (AAD): * + 1-byte Content type and no padding. * AEAD Pre-record nonce: RFC 8446 Section-5.3 * + 8-byte length (64 bits explicit portion added during encryption). * AEAD tag: RFC 5116 Section-5.1 * + 16-byte tag appended to encrypted_record for AEAD_AES_128_GCM ciphertext is exactly 16 octets longer than its corresponding plaintext. Handshake is connection-scoped — negligible RPS impact at this reconnect rate. AWS ACM certificates are free.	None
OAuth 2.0 + JWT — bearer token	Unary interceptor TTL 3600s 1 client(s)	JWT Bearer header RFC 7515 Section-7.1 BASE64URL(UTF8(JWS Protected Header)) ' ' * BASE64URL(JWS Payload) ' ' * BASE64URL(JWS Signature) adds 650 bytes typical due to Payload overhead RFC 7519 §3 and RFC 9101: 650 bytes per message: Algorithm Standard claims RS256 Header 80 Payload (compact/std/ext) 200/400/600 Signature 344	None

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
		Dots 2 Total 626/826/1026 → 650 typical . The token travels in every request's Authorization metadata header. Response messages carry no JWT overhead. Validation enforced by the configured gRPC interceptor.	

⚠️ RPS-IMPACTING — INCREASES NETWORKING COSTS

TACTIC	VALUE	DESCRIPTION	RPS IMPACT
Retry	5% max retries	Retries gRPC packet emission under temporary outages. Increases networking costs as up to N additional packets may be sent per logical request.	+50 req/s



RPS Adjusted by Active Tactics

BASE RPS	EFFECTIVE RPS
1,000	1,050



Security Byte Overhead Applied to Message Sizes

TLS OVERHEAD / MESSAGE	JWT OVERHEAD / REQUEST	EFFECTIVE REQUEST SIZE
30 bytes (typical)	650 bytes (typical)	938 bytes
30 / bytes typical	300 / 650 / 1200 bytes (min/typical/max)	
EFFECTIVE RESPONSE SIZE		
7577 bytes		

MESSAGE TYPE	PROTO SIZE (BYTES)	EFFECTIVE SIZE (BYTES)	REQUESTS / MONTH	THROUGHPUT (GB/MONTH)	TRANSFER COST (USD/MONTH)
Request	258	938	3B	2,377.5369 GB	—
Response	7547	7577	3B	19,205.3273 GB	\$ 1,683.65

📊 Total Cost of Ownership (TCO) Breakdown

Cost Component	Category	Monthly Cost	% of TCO	ROI Impact
AWS Egress (Response Transfer) 1000 eff. RPS · AWS data-out tiers	Networking	\$1683.65	81.0%	▼ Inhibits ROI
mTLS frame overhead (RFC 8446) Included in egress above — no additional charge	Security	\$0 (free)	free	n/a
JWT header overhead (RFC 7519 — request-side) AWS inbound data transfer is free	Security	\$0 (free)	free	n/a
Application Load Balancer (ALB) \$47.45/mo · /api/aws/alb-pricing	Cloud Infra	\$47.45	2.3%	▼ Inhibits ROI
Amazon Macie \$199.00/mo · /api/aws/security-services	Security	\$199.00	9.6%	▼ Inhibits ROI
CloudWatch Logs \$106.00/mo · /api/aws/security-services	Security	\$106.00	5.1%	▼ Inhibits ROI
AWS Audit Manager \$6.00/mo · /api/aws/security-services	Security	\$6.00	0.3%	▼ Inhibits ROI
Containerized Cluster (EKS) \$36.50/mo	Cloud Infra	\$36.50	1.8%	▼ Inhibits ROI
Total Cost of Ownership (TCO)		\$2078.60/mo	100.0%	ROI Efficiency

▲ Promotes ROI = positive efficiency factor. ▼ Inhibits ROI = margin cost (hover for tip). Costs via AWS Pricing API. DR = Disaster Recovery | BC = Business Continuity.

 Unit Economics

TOTAL MONTHLY TCO (NETWORKING + CLOUD INFRA)

\$2078.60

\$1683.65 egress + \$394.95 cloud infra

 Per-Microservice Cost Metrics

TCO / MICROSERVICE / MONTH

\$2078.60

COST / REQUEST

\$0.00000100

2.592.000.000 req/mo

ANNUAL PROJECTION

\$24943.20

12× monthly TCO

This service - full TCO

 Per-End-User Cost Metrics (350 End Users)

TCO / END USER / MONTH

\$5.9389

\$2078.60 ÷ 350 users

TCO / END USER / DAY

\$0.197963

Monthly ÷ 30

BUSINESS USE CASE

BUC1

Pattern cost attribution

 Revenue, ARPU & ROI Analysis

Revenue model: The value entered in Phase 1 is the *expected monthly revenue per end user* for this service. Total monthly revenue = that value × number of end users. **ARPU** = revenue per user per month. **ROI** = (total revenue – TCO) ÷ TCO. **Break-even users** = minimum end users needed to cover full TCO at current ARPU. Net Margin per user = ARPU – TCO per user. Present to Finance and Leadership to frame architecture cost as a revenue-generating investment.

TOTAL MONTHLY
REVENUE

\$2450

\$7.00/user × 350 users

ARPU / MONTH

\$7

Revenue per end user /
month

NET MARGIN / USER /
MO

\$1.0611

ARPU – TCO/user

MONTHLY ROI

17.9%

(Revenue – TCO) ÷ TCO

ANNUAL ROI

17.9%

12× monthly projection

BREAK-EVEN USERS

297

users to cover full TCO

REVENUE / \$1 INFRA

\$1.18

Revenue efficiency

ANNUAL REVENUE

\$29.400

12× monthly revenue

 FinOps Tagging & Labelling Strategy FINOPS BEST PRACTICE

 Tagging and labelling cloud resources is the foundation of cost visibility in FinOps. Without consistent tags, cost allocation across teams, services, and environments is impossible. The following strategy is derived from the **FinOps Foundation Tagging Best Practices** and **AWS Cost Allocation Tags** guidance, applied to the cloud services selected for this microservice.

★ MANDATORY TAGS (APPLY TO ALL SELECTED RESOURCES)

Tag Key	Example Value	Purpose	FinOps Dimension
Environment	production staging dev	Separate cost by environment	Allocation
Team	platform-engineering backend	Charge back costs to the owning team	Chargeback
CostCentre	CC-1042	Finance cost centre mapping	Chargeback
Project	mach-e-commerce-replatform	Group costs by business initiative	Allocation
Application	order-management-service	Identify the owning application	Showback
ManagedBy	terraform cloudformation cdk	Track IaC ownership and drift	Governance
DataClassification	confidential internal public	Support security and compliance policy	Governance

🔧 SERVICE-SPECIFIC TAGS (FOR THIS MICROSERVICE)

Tag Key	Suggested Value	Purpose
ServiceName	order-retriever-system	Exact microservice identifier for cost drill-down
gRPCPattern	unary-rpc	The gRPC streaming pattern used by this service
BaseRPS	1000	Baseline load — supports capacity cost attribution
BusinessUseCase	BUC1	Ties cost to the business capability it supports
ProtocolBuffer	proto3	Serialisation format — relevant to egress cost model
FinOpsReviewed	2026-07-04	Date last reviewed by FinOps/architect for cost fitness

☁️ SELECTED CLOUD SERVICES — TAGGING CHECKLIST

☁️ Application Load Balancer (ALB)

Application

Environment

Team

CostCentre

ServiceName

📌 Tag each ALB individually — one ALB can serve multiple services, so tag at listener/target-group level too.

☁ Security Services (GuardDuty / WAF / KMS / Macie / CloudWatch)

Application Environment Team CostCentre

📌 GuardDuty and Macie are account-level; use AWS Cost Categories for allocation. KMS CMKs: tag per owning service.

☁ Amazon EKS Cluster

Application Environment Team CostCentre ManagedBy

📌 Add tags to the EKS cluster AND all managed node groups. Use AWS Cost Allocation Tags + Kubecost/OpenCost for pod-level attribution.

⚠ **Governance note:** Tags are only effective when enforced via **AWS Tag Policies** (AWS Organizations) and monitored through **AWS Cost Explorer Tag Coverage** reports. Untagged resources should trigger automated alerts via **AWS Config Rules** (e.g. `required-tags`). Discuss tagging governance with your FinOps and Cloud Centre of Excellence (CCoE) team.

🏠 FinOps Architecture Notes & Disclaimers

🏠 1 — IP Address Planning (CIDR Sizing)

The more redundancy you build into AWS (Multi-AZ, replicas, NAT Gateways, ENIs), the more private IP addresses are consumed. Plan your VPC CIDR block generously (e.g. /16 = 65,536 IPs) and **leave enough address space within each subnet CIDR for auto-scaling headroom**. A /24 subnet provides only 251 usable IPs — exhausted quickly in a multi-tier, auto-scaled architecture.

🏠 2 — Multi-Tier Architecture in AWS Terms

Separating MACH components (frontend, API gateway, gRPC services, databases) into **different subnets** makes this a **multi-tier architecture** in AWS terms. Each tier needs dedicated subnets with its own Security Groups and NACLs.

🏠 3 — Intra-VPC Connectivity

Use **VPC Endpoints** (PrivateLink) to access AWS services without internet routing; **Transit Gateway** for cross-VPC connectivity; **VPC Flow Logs** for visibility. Each component in a separate subnet requires explicit route table entries.

* Outbound (response) data transfer cost uses AWS standard internet egress tiered pricing for US East (N. Virginia): first 10 TB at \$0.09/GB, next 40 TB at \$0.085/GB, next 100 TB at \$0.07/GB, beyond 150 TB at \$0.05/GB.

Inbound (request) traffic is free. Effective sizes include TLS (RFC 8446) and JWT (RFC 7519) overhead where applicable. [AWS Pricing Calculator](#)

Phase 5 — Architecture Portfolio Summary



Final phase: aggregated TCO across all modelled microservices. Download the PDF to share with stakeholders.

Each service you calculate and add to the portfolio appears here with its description, Business Use Case, and total monthly TCO. Use **Add to Portfolio** in Phase 4. Data is stored in your browser's **local storage** — no server required, survives page reload.

Aggregated Unit Economics

TOTAL ARCHITECTURE TCO \$2078.60/mo sum of 1 services	TOTAL ANNUAL TCO \$24943.20/yr 12 × monthly	AVG TCO / MICROSERVICE \$2078.60/mo mean
TOTAL RPS 1000 aggregate req/s	COST / REQUEST \$0.000001 avg	COST / USER / MONTH \$5.9389 350 users

Portfolio Revenue, ARPU & ROI Analysis

Aggregate ROI across all modelled microservices. Set Expected Revenue per Transaction in Phase 1 per service. **Break-even** = users to cover full portfolio TCO. Share with Finance and Leadership.

TOTAL MONTHLY REVENUE \$2450.00 across all services	TOTAL ANNUAL REVENUE \$29400.00 12 × monthly	PORTFOLIO ARPU \$7.0000/user/mo weighted avg
MONTHLY ROI 17.9% (rev - TCO) ÷ TCO	ANNUAL ROI 17.9% 12× monthly	NET MONTHLY PROFIT \$371.40 rev - TCO
NET ANNUAL PROFIT \$4456.80 12×	BREAK-EVEN USERS 297 to cover portfolio TCO	REVENUE / \$1 INFRA \$1.18 efficiency

MACH Architecture — TCO Portfolio Summary

Generated: 3/7/2026, 22:49:29

Service	BUC	Purpose	Monthly TCO	Annual TCO
order-retriever-system	BUC1	pnomom	\$2078.60/mo	\$24943.20/yr
TOTAL (1 services)			\$2078.60/mo	\$24943.20/yr

Architectural TCO estimate. Validate with Engineering, FinOps, Finance and Procurement before acting on these figures.